Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/19/2007
PI Name:	Liu, Feng-chuan Ph.D.		
Project Title:	Critical Thermal Transport in a Cross-Over Range from 3D to 2D behavior		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	818-354-1706
Organization Name:	JPL		
PI Address 1:	4800 Oak grove Drive, MS 264-723		
PI Address 2:	California Institute of Technology		
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91109	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	10/01/1998	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 09/30/2007 per D. Strayer (1/07) Note - Strayer says changed to ground in FY 2004		
Key Personnel Changes/Previous PI:	liu, feng-chuan		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This task is part of the Boundary Effects on the Superfluid Tansition (BEST) Experiment previously scheduled to fly on the M2 MIssion of the Low Temperature Microgravity Physics Facility (LTMPF) on the International Space Station. BEST Experiment was returned to ground investigation in 2002 due to LTMPF funding reduction and re-plan. The science investigation studies the effects of a solid bounday on a fluid system near its critical transition, or specifically the superfluid transition of helium-4. The investigation consists of both measurement that can be done in the ground laboratory and measurement that can only be achieved in a microgravity environment in space. To perform systematic measurement on the thermal conductivity of helium-4 near its superfluid transition either as a 3-dimentional sample or in a 2-dimensional confinement. Comparing the results between the two boundary conditions will give tell us the effects and dimentionality and solid boundary on the superfluid transition. Confinement of different sizes willalso enabe us to test the hypothesis of dynamic finite size scaling theory. The task address two fundamental questions in physics: how does a solid boundary affect the critical phenomena in a fluid system, and does the finite size scaling work for dyanmic behaviors. Neither questions are properly understaood at this point because the investigation involves several conditions at the same time: a well defined confinement media, sub-nano-kelvin temperature and thermal contraol, as well as microgravity environment . With the microgravity environment on ISS and the stable low temparture environment provided by LTMPF, we have all the conditions to adress these important issues.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During FY 2006, our effort was focused on data analysis and publication. We analyzed the thermal conductivity data for liquid helium confined in 1µmx10µm rectangular Microchannel plate (1mm thick). We prepared and submitted a manuscript for refereed publication in the Journal of Low Temperature Physics (JLTP). The reports from the referees were all positive and recommended publication after a few minor clarifications/modifications. We expect the paper will appear in the early 2007 issue.
Bibliography Type:	Description: (Last Updated: 04/08/2008)
Articles in Peer-reviewed Journals	Jerebets S, Liu Y, Liu F, Ahlers G. "Finite-Size Effects on the Thermal Resistivity of 4He near the Superfluid Transition." Journal of Low Temperature Physics. in press, 2007. , Jan-2007